(a) Exchange the physical range

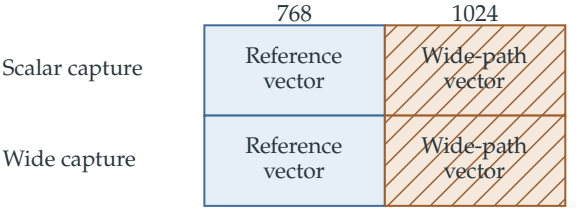
Native AV replay; q = 1 on both sides

Scalar capture → Zero-pad P; use 1024-entry V

Wide capture → Crop P and V to 768 entries

Visible probabilities remain unchanged; physical extent is exchanged.

(b) The full AV result follows the range



Exact AV output; state is a separate obligation.